

FORMULACIÓN DIRECTA

A **B**

EST) $6 x_1 + 16 x_2 \leq 48000$

SOL) $12 x_1 + 6 x_2 \leq 42000$

PIN) $9 x_1 + 9 x_2 \leq 36000$

$$Z = 4 x_1 + 3 x_2 \Rightarrow \text{Máx}$$

$$x_1 , \quad x_2 \geq 0$$

FORMULACIÓN DUAL

EST

SOL

PIN

A)

B)

FORMULACIÓN DIRECTA

A **B**

EST) $6 x_1 + 16 x_2 \leq 48000$

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$Z = 4 x_1 + 3 x_2 \Rightarrow \text{Máx}$

$x_1 , \quad x_2 \geq 0$

FORMULACIÓN DUAL

EST

SOL

PIN

A)

$$6 y_1 + 12 y_2 + 9 y_3 \geq 4$$

B)

FORMULACIÓN DIRECTA

A

B

EST) $6 x_1 + 16 x_2 \leq 48000$

SOL) $12 x_1 + 6 x_2 \leq 42000$

PIN) $9 x_1 + 9 x_2 \leq 36000$

$Z = 4 x_1 + 3 x_2 \Rightarrow \text{Máx}$

$x_1 , \quad x_2 \geq 0$

FORMULACIÓN DUAL

EST

SOL

PIN

A) $6 y_1 + 12 y_2 + 9 y_3 \geq 4$

B) $16 y_1 + 6 y_2 + 9 y_3 \geq 3$

$Z = 48.000 y_1 + 42.000 y_2 + 36.000 y_3 \Rightarrow \text{Mín}$

$y_1, \quad y_2, \quad y_3 \geq 0$

FORMULACIÓN DIRECTA

A **B**

EST) $6 x_1 + 16 x_2 \leq 48000$

SOL) $12 x_1 + 6 x_2 \leq 42000$

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$Z = 4 x_1 + 3 x_2 \Rightarrow \text{Máx}$

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FORMULACIÓN DUAL

EST

SOL

PIN

$$\text{A)} \quad 6 y_1 + 12 y_2 + 9 y_3 \geq 4$$

$$\text{B)} \quad 16 y_1 + 6 y_2 + 9 y_3 \geq 3$$

$$Z = 48.000 y_1 + 42.000 y_2 + 36.000 y_3 \Rightarrow \text{Mín}$$

FORMULACIÓN DUAL

EST

SOL

PIN

$$\text{A)} \quad 6 y_1 + 12 y_2 + 9 y_3 \geq 4$$

$$\text{B)} \quad 16 y_1 + 6 y_2 + 9 y_3 \geq 3$$

$$Z = 48.000 y_1 + 42.000 y_2 + 36.000 y_3 \Rightarrow \text{Mín}$$

$$y_i \geq 0$$

DIRECTO

$$\text{MAX: } Z = 4 x_1 + 3 x_2$$

$$6 x_1 + 16 x_2 \leq 48000$$

$$12 x_1 + 6 x_2 \leq 42000$$

$$9 x_1 + 9 x_2 \leq 36000$$

$$x_i \geq 0$$

DUAL

$$\text{MIN: } 48.000 y_1 + 42.000 y_2 + 36.000 y_3$$

$$6 y_1 + 12 y_2 + 9 y_3 \geq 4$$

$$16 y_1 + 6 y_2 + 9 y_3 \geq 3$$

$$y_i \geq 0$$

EST SOL PIN A B

A) 6 y_1 + 12 y_2 + 9 y_3 - y_4 = 4

B) 16 y_1 + 6 y_2 + 9 y_3 - y_5 = 3

$Z = 48.000 y_1 + 42.000 y_2 + 36.000 y_3 \Rightarrow \text{Mín}$

$y_i \geq 0$

		c_j	48000	42000	36000	0	0	M	M	
c_k	x_k	B_k	A'_1	A'_2	A'_3	A'_4	A'_5	$A'_{\mu 1}$	$A'_{\mu 2}$	b_i/a_{ij}
M	y_4	4	6	12	9	-1		1		0.6667
M	y_5	3	16	6	9	-1	-1		1	0.1875
$Z = 7 M$			22M -48000	18M -42000	18M -36000	-M	-M	0	0	$z_j - c_j$

c_k	x_k	B_k	A'_1	A'_2	A'_3	A'_4	A'_5	$A'_{\mu 1}$	$A'_{\mu 2}$	b_i/a_{ij}
M	y_4	2.875		9.75	5.625	-1	-0.375		-0.375	0.2949
48000	y_1	0.1875	1	0.375	0.5625	-1	-0.0625	1	0.0625	0.5
$Z = 2.875 M + 9000$			0	9.75M -24000	5.625M -9000	-M	-0.375M +3000	0	-1.375M +3000	$z_j - c_j$

c_k	x_k	B_k	A'_1	A'_2	A'_3	A'_4	A'_5	$A'_{\mu 1}$	$A'_{\mu 2}$	b_i/a_{ij}
42000	y_2	0.2949		1	0.5769	-0.1026	-0.0384	0.1026	-0.375	0.5111
48000	y_1	0.0769	1		0.3462	0.0384	-0.0769	-0.0384	0.0625	0.2222
$Z = 16076,92$			0	0	4846.15	-2461.54	-2076.92	2466-M	-12750 -M	$z_j - c_j$

c_k	x_k	B_k	A'_1	A'_2	A'_3	A'_4	A'_5	$A'_{\mu 1}$	$A'_{\mu 2}$
42000	y_2	0.1667	-1.6667	1		-0.1667	0.1667	0.1667	-0.1667
36000	y_3	0.2222	2.8889		1	0.1111	-0.2222	-0.1111	0.2222
$Z = 15000$			-14000	0	0	-3000	-1000	3000-M	1000-M

TEOREMA FUNDAMENTAL DE LA DUALIDAD

DIRECTA FINITA $\Leftrightarrow$ DUAL FINITA

$$Z_{\text{ÓPTIMO DIRECTO}} = Z_{\text{ÓPTIMO DUAL}}$$

DEGENERADA $\Leftrightarrow$ ALTERNATIVA

INCOMPATIBLE $\Leftrightarrow$ POL. ABIERTO

PROBLEMA DIRECTO

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graph TD; A[PROBLEMA DIRECTO] --> B[FORMA CANÓNICA]; A --> C[FORMA ESTÁNDAR]; B --> D[TRANSFORMACIÓN SIMÉTRICA]; C --> E[TRANSFORMACIÓN ASIMÉTRICA]; D --> F[PROBLEMA DUAL]; E --> F;
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FORMA CANÓNICA

FORMA ESTÁNDAR

TRANSFORMACIÓN
SIMÉTRICA

TRANSFORMACIÓN
ASIMÉTRICA

PROBLEMA DUAL

DUAL SIMÉTRICO

DIRECTO	DUAL
$A \cdot X \leq B$ $X \geq 0$ $Z = C \cdot X \rightarrow \text{Máx}$	

DUAL SIMÉTRICO

DIRECTO	DUAL
$A \cdot X \leq B$ $X \geq 0$ $Z = C \cdot X \rightarrow \text{Máx}$	$A^T \cdot Y \geq C$ $Y \geq 0$ $Z = B \cdot Y \rightarrow \text{Mín}$

DUAL SIMÉTRICO

DIRECTO	DUAL
$A \cdot X \leq B$ $X \geq 0$ $Z = C \cdot X \rightarrow \text{Máx}$	$A^T \cdot Y \geq C$ $Y \geq 0$ $Z = B \cdot Y \rightarrow \text{Mín}$
$A \cdot X \geq B$ $X \geq 0$ $Z = C \cdot X \rightarrow \text{Mín}$	

DUAL SIMÉTRICO

DIRECTO	DUAL
$A \cdot X \leq B$ $X \geq 0$ $Z = C \cdot X \rightarrow \text{Máx}$	$A^T \cdot Y \geq C$ $Y \geq 0$ $Z = B \cdot Y \rightarrow \text{Mín}$
$A \cdot X \geq B$ $X \geq 0$ $Z = C \cdot X \rightarrow \text{Mín}$	$A^T \cdot Y \leq C$ $Y \geq 0$ $Z = B \cdot Y \rightarrow \text{Máx}$

DUAL ASIMÉTRICO

DIRECTO	DUAL
$A \cdot X = B$ $X \geq 0$ $Z = C \cdot X \rightarrow \text{Máx}$	

DUAL ASIMÉTRICO

DIRECTO	DUAL
$A \cdot X = B$ $X \geq 0$ $Z = C \cdot X \rightarrow \text{Máx}$	$A^T \cdot Y \geq C$ $Z = B \cdot Y \rightarrow \text{Mín}$

DUAL ASIMÉTRICO

DIRECTO	DUAL
$A \cdot X = B$ $X \geq 0$ $Z = C \cdot X \rightarrow \text{Máx}$	$A^T \cdot Y \geq C$ $Z = B \cdot Y \rightarrow \text{Mín}$
$A \cdot X = B$ $X \geq 0$ $Z = C \cdot X \rightarrow \text{Mín}$	

DUAL ASIMÉTRICO

DIRECTO	DUAL
$A \cdot X = B$ $X \geq 0$ $Z = C \cdot X \rightarrow \text{Máx}$	$A^T \cdot Y \geq C$ $Z = B \cdot Y \rightarrow \text{Mín}$
$A \cdot X = B$ $X \geq 0$ $Z = C \cdot X \rightarrow \text{Mín}$	$A^T \cdot Y \leq C$ $Z = B \cdot Y \rightarrow \text{Máx}$

DUAL SIMÉTRICO

MAX: $Z = 4 x_1 + 3 x_2 - x_3$

$$\left\{ \begin{array}{l} 16 x_1 + 6 x_2 + 2 x_3 \leq 480 \\ 4 x_1 + 5 x_2 - 7 x_3 \geq 100 \\ 10 x_1 + 8 x_2 - 3 x_3 \leq 500 \end{array} \right.$$

$$x_i \geq 0$$

$$\text{MAX: } Z = 4 x_1 + 3 x_2 - x_3$$

$$\left\{ \begin{array}{l} 16 x_1 + 6 x_2 + 2 x_3 \leq 480 \\ -4 x_1 - 5 x_2 + 7 x_3 \leq -100 \\ 10 x_1 + 8 x_2 - 3 x_3 \leq 500 \end{array} \right.$$

$$x_i \geq 0$$

**DIRECTO
FORMA
CANÓNICA**

$$\text{MAX: } Z = 4 x_1 + 3 x_2 - x_3$$

$$\begin{cases} 16 x_1 + 6 x_2 + 2 x_3 \leq 480 \\ -4 x_1 - 5 x_2 + 7 x_3 \leq -100 \\ 10 x_1 + 8 x_2 - 3 x_3 \leq 500 \end{cases}$$

$$x_i \geq 0$$

$$\text{MIN: } Z = 480 y_1 - 100 y_2 + 500 y_3$$

$$\begin{cases} 16 y_1 - 4 y_2 + 10 y_3 \geq 4 \\ 6 y_1 - 5 y_2 + 8 y_3 \geq 3 \\ 2 y_1 + 7 y_2 - 3 y_3 \geq -1 \end{cases}$$

$$y_i \geq 0$$

DIRECTO

$$\text{MAX: } Z = 4 x_1 + 3 x_2 - x_3$$

$$\begin{cases} 16 x_1 + 6 x_2 + 2 x_3 \leq 480 \\ 4 x_1 + 5 x_2 - 7 x_3 \geq 100 \\ 10 x_1 + 8 x_2 - 3 x_3 \leq 500 \end{cases}$$

$$x_i \geq 0$$

DUAL

$$\text{MIN: } Z = 480 y_1 - 100 y_2 + 500 y_3$$

$$\begin{cases} 16 y_1 - 4 y_2 + 10 y_3 \geq 4 \\ 6 y_1 - 5 y_2 + 8 y_3 \geq 3 \\ -2 y_1 - 7 y_2 + 3 y_3 \leq 1 \end{cases}$$

$$y_i \geq 0$$

$$\text{MAX: } Z = 4 x_1 + 3 x_2 - x_3$$

$$\left\{ \begin{array}{l} 16 x_1 + 6 x_2 + 2 x_3 \leq 480 \\ 4 x_1 + 5 x_2 - 7 x_3 = 100 \\ 10 x_1 + 8 x_2 - 3 x_3 \leq 500 \end{array} \right.$$

$$x_i \geq 0$$

MAX: $Z = 4 x_1 + 3 x_2 - x_3$

$$\left\{ \begin{array}{l} 16 x_1 + 6 x_2 + 2 x_3 \leq 480 \\ 4 x_1 + 5 x_2 - 7 x_3 \leq 100 \\ 4 x_1 + 5 x_2 - 7 x_3 \geq 100 \\ 10 x_1 + 8 x_2 - 3 x_3 \leq 500 \end{array} \right.$$

$$x_i \geq 0$$

MAX: $Z = 4 x_1 + 3 x_2 - x_3$

$$\left\{ \begin{array}{l} 16 x_1 + 6 x_2 + 2 x_3 \leq 480 \\ 4 x_1 + 5 x_2 - 7 x_3 \leq 100 \\ -4 x_1 - 5 x_2 + 7 x_3 \leq -100 \\ 10 x_1 + 8 x_2 - 3 x_3 \leq 500 \end{array} \right.$$

$$x_i \geq 0$$

**DIRECTO
FORMA
CANÓNICA**

MAX: $Z = 4 x_1 + 3 x_2 - x_3$

$$\left\{ \begin{array}{l} 16 x_1 + 6 x_2 + 2 x_3 \leq 480 \\ 4 x_1 + 5 x_2 - 7 x_3 \leq 100 \\ -4 x_1 - 5 x_2 + 7 x_3 \leq -100 \\ 10 x_1 + 8 x_2 - 3 x_3 \leq 500 \end{array} \right.$$

$x_i \geq 0$

MIN: $Z = 480 y_1 + 100 y_2 - 100 y_2' + 500 y_3$

$$\left\{ \begin{array}{l} 16 y_1 + 4 y_2 - 4 y_2' + 10 y_3 \geq 4 \\ 6 y_1 + 5 y_2 - 5 y_2' + 8 y_3 \geq 3 \\ 2 y_1 - 7 y_2 + 7 y_2' - 3 y_3 \geq -1 \end{array} \right.$$

$y_i \geq 0$

DIRECTO

MAX: $Z = 4 x_1 + 3 x_2 - x_3$

$$\left\{ \begin{array}{l} 16 x_1 + 6 x_2 + 2 x_3 \leq 480 \\ 4 x_1 + 5 x_2 - 7 x_3 = 100 \\ 10 x_1 + 8 x_2 - 3 x_3 \leq 500 \end{array} \right.$$

$$x_i \geq 0$$

DUAL

MIN: $Z = 480 y_1 + 100 y_2 - 100 y_2' + 500 y_3$

$$\left\{ \begin{array}{l} 16 y_1 + 4 y_2 - 4 y_2' + 10 y_3 \geq 4 \\ 6 y_1 + 5 y_2 - 5 y_2' + 8 y_3 \geq 3 \\ -2 y_1 + 7 y_2 - 7 y_2' + 3 y_3 \leq 1 \end{array} \right.$$

$$y_i \geq 0$$

DUAL ASIMÉTRICO

$$\text{MAX: } Z = 4 x_1 + 3 x_2 - x_3$$

$$\left\{ \begin{array}{l} 16 x_1 + 6 x_2 + 2 x_3 \leq 480 \\ 4 x_1 + 5 x_2 - 7 x_3 = 100 \\ 10 x_1 + 8 x_2 - 3 x_3 \leq 500 \end{array} \right.$$

$$x_i \geq 0$$

FORMA ESTÁNDAR

$$\text{MAX: } Z = 4 x_1 + 3 x_2 - x_3$$

$$\left\{ \begin{array}{l} 16 x_1 + 6 x_2 + 2 x_3 + x_4 = 480 \\ 4 x_1 + 5 x_2 - 7 x_3 = 100 \\ 10 x_1 + 8 x_2 - 3 x_3 + x_5 = 500 \end{array} \right.$$

$$x_i \geq 0$$

DIRECTO

$$\text{MAX: } Z = 4 x_1 + 3 x_2 - x_3$$

$$\left\{ \begin{array}{l} 16 x_1 + 6 x_2 + 2 x_3 \leq 480 \\ 4 x_1 + 5 x_2 - 7 x_3 = 100 \\ 10 x_1 + 8 x_2 - 3 x_3 \leq 500 \end{array} \right.$$

$$x_i \geq 0$$

DUAL

$$\text{MIN: } Z = 480 y_1 + 100 y_2 + 500 y_3$$

$$\left\{ \begin{array}{l} 16 y_1 + 4 y_2 + 10 y_3 \geq 4 \\ 6 y_1 + 5 y_2 + 8 y_3 \geq 3 \\ 2 y_1 - 7 y_2 - 7 y_3 \geq -1 \\ y_1 \geq 0 \\ y_3 \geq 0 \end{array} \right.$$

PASAJE DE TABLA ÓPTIMA DIRECTA A TABLA ÓPTIMA DUAL

(Y VICEVERSA)

MAX: $Z = 4 x_1 + 3 x_2$

Sujeto a:

$$\left\{ \begin{array}{l} 6 x_1 + 16 x_2 \leq 48000 \\ 12 x_1 + 6 x_2 \leq 42000 \\ 9 x_1 + 9 x_2 \leq 36000 \end{array} \right.$$

siendo: $x_1, x_2 \geq 0$ y continuas

0	x_3	14.000			1	5/3	-26/9
4	x_1	3.000	1			1/6	-1/9
3	x_2	1.000		1		-1/6	2/9
Z = 15.000			0	0	0	1/6	2/9
			y_4	y_5	y_1	y_2	y_3

0	x_3	14.000			1	5/3	-26/9
4	x_1	3.000	1			1/6	-1/9
3	x_2	1.000		1		-1/6	2/9
Z = 15.000			0	0	0	1/6	2/9
			y_4	y_5	y_1	y_2	y_3



Z = 15.000							

0	x_3	14.000			1	5/3	-26/9
4	x_1	3.000	1			1/6	-1/9
3	x_2	1.000		1		-1/6	2/9
Z = 15.000			0	0	0	1/6	2/9
			y_4	y_5	y_1	y_2	y_3

	y_2						
	y_3						
Z = 15.000							

0	x_3	14.000			1	5/3	-26/9
4	x_1	3.000	1			1/6	-1/9
3	x_2	1.000		1		-1/6	2/9
Z = 15.000			0	0	0	1/6	2/9

y_4 y_5 y_1 y_2 y_3

42.000	y_2	1/6					
36.000	y_3	2/9					
Z = 15.000							

0	x_3	14.000			1	5/3	-26/9
4	x_1	3.000	1			1/6	-1/9
3	x_2	1.000		1		-1/6	2/9
Z = 15.000			0	0	0	1/6	2/9
			y_4	y_5	y_1	y_2	y_3

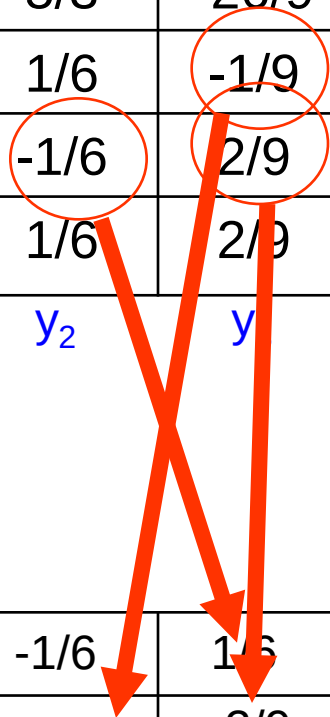
42.000	y_2	1/6		1			
36.000	y_3	2/9			1		
Z = 15.000				0	0		
			x_3	x_4	x_5	x_1	x_2

0	x_3	14.000			1	$5/3$	$-26/9$
4	x_1	3.000	1			$1/6$	$-1/9$
3	x_2	1.000		1		$-1/6$	$2/9$
$Z = 15.000$			0	0	0	$1/6$	$2/9$
			y_4	y_5	y_1	y_2	y_3

42.000	y_2	$1/6$	$-5/3$	1		$-1/6$	
36.000	y_3	$2/9$	$26/9$		1		
$Z = 15.000$				0	0		
			x_3	x_4	x_5	x_1	x_2

0	x_3	14.000			1	$5/3$	$-26/9$
4	x_1	3.000	1			$1/6$	$-1/9$
3	x_2	1.000		1		$-1/6$	$2/9$
$Z = 15.000$			0	0	0	$1/6$	$2/9$
			y_4	y_5	y_1	y_2	y_3

42.000	y_2	$1/6$	$-5/3$	1		$-1/6$	$1/6$
36.000	y_3	$2/9$	$26/9$		1	$1/9$	$-2/9$
$Z = 15.000$				0	0		
			x_3	x_4	x_5	x_1	x_2

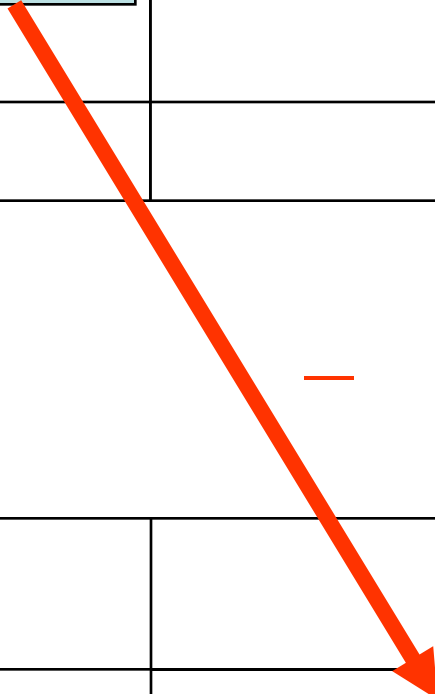


0	x_3	14.000			1	5/3	-26/9
4	x_1	3.000	1			1/6	-1/9
3	x_2	1.000		1		-1/6	2/9
Z = 15.000			0	0	0	1/6	2/9
			y_4	y_5	y_1	y_2	y_3

42.000	y_2	1/6	5/3	1		-1/6	1/6
36.000	y_3	2/9	26/9		1	1/9	-2/9
Z = 15.000			-14.000	0	0	-3.000	-1.000
			x_3	x_4	x_5	x_1	x_2

	x_i		
$Z \Rightarrow \text{MAX}$			

	y_j		
$Z \Rightarrow \text{MIN}$			



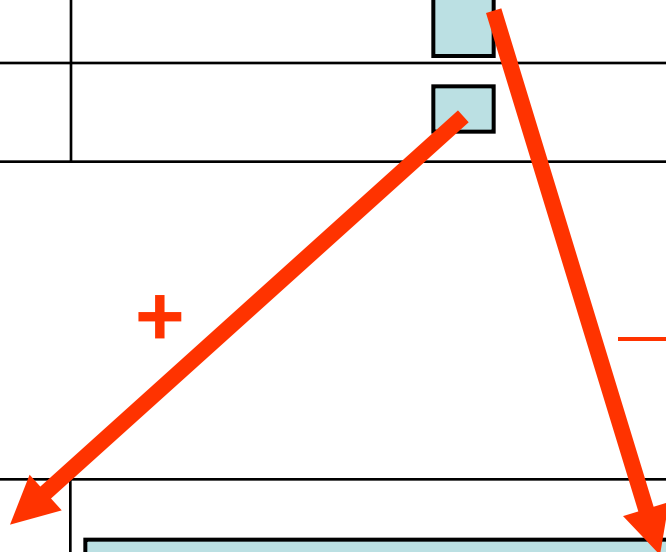
	x_i		
$Z \Rightarrow \text{MIN}$			

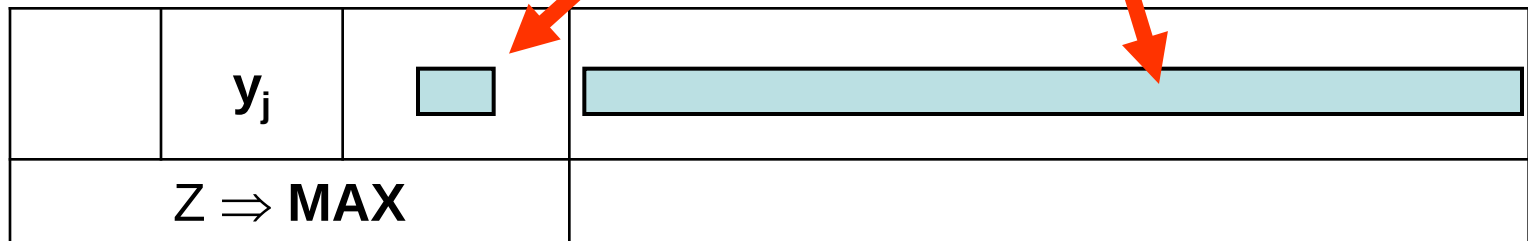
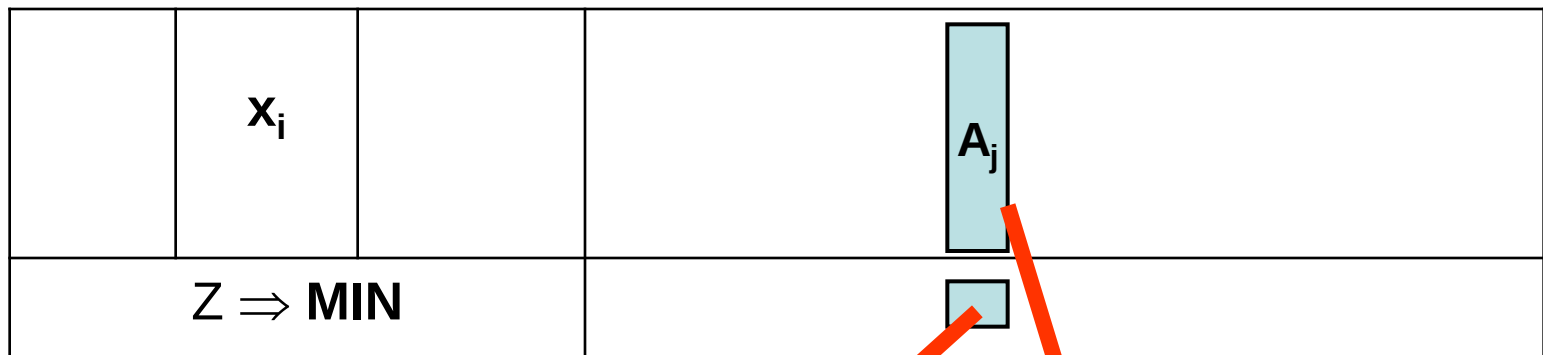
+

	y_j		
$Z \Rightarrow \text{MAX}$			

	x_i		A_j	
$Z \Rightarrow \text{MAX}$				

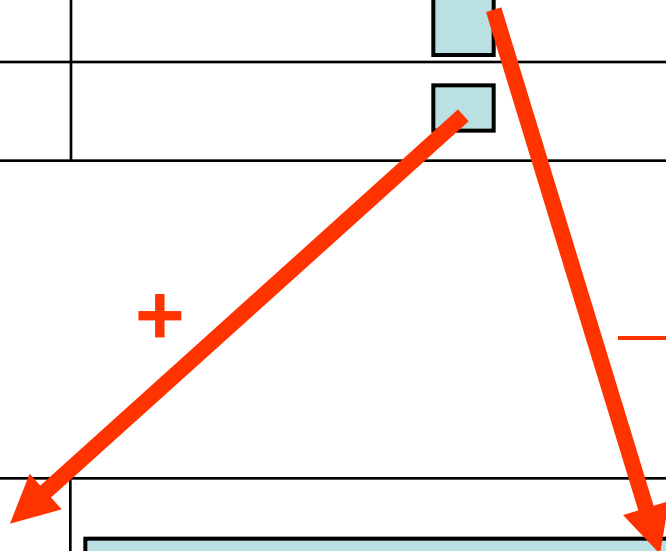
	y_j		
$Z \Rightarrow \text{MIN}$			





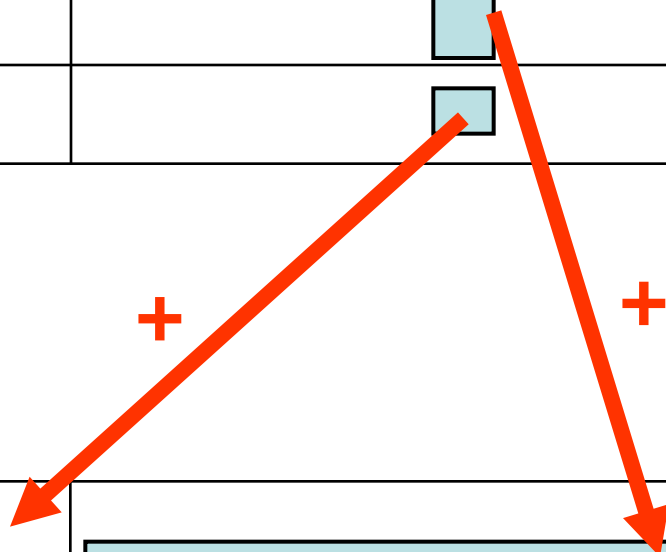
	x_i		λ_j	
$Z \Rightarrow \text{MAX}$				

	y_j		
$Z \Rightarrow \text{MIN}$			



	x_i		λ_j	
$Z \Rightarrow \text{MIN}$				

	y_j		
$Z \Rightarrow \text{MAX}$			



MIN: $Z = x_1 + 0,5 x_2 + 2 x_3$

Sujeto a:

$$\left\{ \begin{array}{l} 3 x_1 + 2 x_2 + 4 x_3 \leq 3,6 \\ x_1 + 0,5 x_2 + 2 x_3 \geq 1,2 \\ x_1 + x_2 + x_3 = 1 \end{array} \right.$$

siendo: $x_1, x_2, x_3 \geq 0$

			1	0,5	2			M
c_K	x_K	B_K	A_1	A_2	A_3	A_4	A_5	λ
1	x_1	0,4	1	2		1		4
0	x_5	0,4		-0,5		-1	1	-2
2	x_3	0,6		1	1	-1		-3
Z (MIN) = 1,6				-0,5		-1		-2-M

 y_4
 y_5
 y_6
 y_1
 y_2
 y_3

			3,6	1,2	1			
b_K	x_K	B_K	A'_1	A'_2	A'_3	A'_4	A'_5	A'_6
3,6	y_1	1	1	1		-1		1
1	y_3	-2		-2	1	4		-3
0	y_5	0,5		0,5		-2	1	-1
Z (MAX) = 1,6				0,4		0,4		0,6

 x_4
 x_5
 x_6
 x_1
 x_2
 x_3

			1	0,5	2			M
c_K	x_K	B_K	A₁	A₂	A₃	A₄	A₅	λ
1	x ₁	0,4	1	2		1		4
0	x ₅	0,4		-0,5		-1	1	-2
2	x ₃	0,6		1	1	-1		-3
Z (MIN) = 1,6				-0,5		-1		-2-M
			y ₄	y ₅	y ₆	y ₁	y ₂	y ₃

			3,6	1,2	1			
b_K	x_K	B_K	A'₁	A'₂	A'₃	A'₄	A'₅	A'₆
3,6	y ₁	1	1	1		-1		1
1	y ₃	-2		-2	1	4		3
0	y ₅	0,5		0,5		-2	1	-1
Z (MAX) = 1,6				0,4		0,4		0,6
			x ₄	x ₅	x ₆	x ₁	x ₂	x ₃

			1	0,5	2			M
c_K	x_K	B_K	A_1	A_2	A_3	A_4	A_5	λ
1	x_1	0,4	1	2		1		4
0	x_5	0,4		-0,5		-1	1	-2
2	x_3	0,6		1	1	-1		-3
Z (MIN) = 1,6				-0,5		-1		-2-M

 y_4
 y_5
 y_6
 y_1
 y_2
 y_3

			3,6	1,2	1			
b_K	x_K	B_K	A'_1	A'_2	A'_3	A'_4	A'_5	A'_6
3,6	y_1	1	1	1		-1		1
1	y_3	-2		-2	1	4		3
0	y_5	0,5		0,5		-2	1	-1
Z (MAX) = 1,6				0,4		0,4		0,6

 x_4
 x_5
 x_6
 x_1
 x_2
 x_3

			1	0,5	2			M
c_K	x_K	B_K	A_1	A_2	A_3	A_4	A_5	λ
1	x_1	0,4	1	2		1		4
0	x_5	0,4		-0,5		-1	1	-2
2	x_3	0,6		1	1	-1		-3
Z (MIN) = 1,6				-0,5		-1		-2-M

 y_4
 y_5
 y_6
 y_1
 y_2
 y_3

			3,6	1,2	1			
b_K	x_K	B_K	A'_1	A'_2	A'_3	A'_4	A'_5	A'_6
3,6	y_1	1	1	1		-1		1
1	y_3	-2		-2	1	4		-3
0	y_5	0,5		0,5		-2	1	-1
Z (MAX) = 1,6				0,4		0,4		0,6

 x_4
 x_5
 x_6
 x_1
 x_2
 x_3

			1	0,5	2			M
c_K	x_K	B_K	A_1	A_2	A_3	A_4	A_5	λ
1	x_1	0,4	1	2		1		4
0	x_5	0,4		-0,5		-1	1	-2
2	x_3	0,6		1	1	-1		-3
Z (MIN) = 1,6				-0,5		-1		-2-M

 y_4
 y_5
 y_6
 y_1
 y_2
 y_3

			3,6	1,2	1			
b_K	x_K	B_K	A'_1	A'_2	A'_3	A'_4	A'_5	A'_6
3,6	y_1	1	1	1		-1		1
1	y_3	-2		-2	1	4		3
0	y_5	0,5		0,5		-2	1	-1
Z (MAX) = 1,6				0,4		0,4		0,6

 x_4
 x_5
 x_6
 x_1
 x_2
 x_3

TEOREMA FUNDAMENTAL DE LA DUALIDAD

DIRECTA FINITA $\Leftrightarrow$ DUAL FINITA

$$Z_{\text{ÓPTIMO DIRECTO}} = Z_{\text{ÓPTIMO DUAL}}$$

DEGENERADA $\Leftrightarrow$ ALTERNATIVA

INCOMPATIBLE $\Leftrightarrow$ POL. ABIERTO

		c_j	3	3	0	0	0
c_k	x_k	B	A_1	A_2	A_3	A_4	A_5
0	x_3	14.000			1	5/3	-26/9
3	x_1	3.000	1			1/6	-1/9
3	x_2	1.000		1		-1/6	2/9
Z = 12.000			0	0	0	0*	3/9

ALTERNATIVA

		c_j	3	3	0	0	0
c_k	x_k	B	A_1	A_2	A_3	A_4	A_5
42.000	y_2	0	-5/3	1		-1/6	1/6
36.000	y_3	3/9	26/9		1	1/9	-2/9
Z = 12.000			-14.000	0	0	-3.000	-1.000

DEGENERADA

INTERPRETACIÓN ECONÓMICA DEL PROBLEMA DUAL

6 y_1

$$6 y_1 + 12 y_2$$

$$6 y_1 + 12 y_2 + 9 y_3$$

$$6 y_1 + 12 y_2 + 9 y_3 \geq 4$$

EST

SOL

PIN

A)

6 y_1 +

12 y_2 +

9 y_3

$\geq$

4

EST

SOL

PIN

A) $6 y_1 + 12 y_2 + 9 y_3 \geq 4$

B) $16 y_1 + 6 y_2 + 9 y_3 \geq 3$

EST

SOL

PIN

A) $6 y_1 + 12 y_2 + 9 y_3 \geq 4$

B) $16 y_1 + 6 y_2 + 9 y_3 \geq 3$

$Z = 48.000 y_1 + 42.000 y_2 + 36.000 y_3 \Rightarrow \text{Mín}$

	EST	SOL	PIN	A	B
--	-----	-----	-----	---	---

A) $6 y_1 + 12 y_2 + 9 y_3 - y_4 = 4$

B) $16 y_1 + 6 y_2 + 9 y_3 - y_5 = 3$

$Z = 48.000 y_1 + 42.000 y_2 + 36.000 y_3 \Rightarrow \text{Mín}$